MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2026

Fourth Semester

FUNCTIONAL ZOOLOGY

DSC-B – MG4DSCZGY202

Duration: 1.5 hrs. Maximum Marks: 50

PART A

Multiple Choice Questions. Answer all questions.

Each question carries 1 mark. (12 x 1 = 12 marks)

1. Deficiency of vitamin B_1 results in [U] [CO:1]

   a) Beriberi

   b) Scurvy

   c) Dermatitis

   d) Pellagra

2. ---------- is a neurotransmitter [U] [CO:2]

   a) Thiamine

   b) Acetylcholine

   c) Carbon monoxide

   d) Sodium ions

3. Hypoxia is defined as [U] [CO:1]

   a) Deficiency of CO_2 in blood

   b) Breathlessness

c) Deficiency of O_2 in tissues

d) Retention of high concentration of CO_2 in blood

4. In infancy hypothyroidism results in [U] [CO:3]

a) Simple Goitre

b) Myxoedema

c) Cretinism

d) Exophthalmic goitre

5. Immunity developed in the body after exposure to an antigen is called [U] [CO:4]

a) Passive immunity

b) Innate immunity

c) Active immunity

d) Natural immunity

6. Name the vitamin involved in blood coagulation [U] [CO:1]

a) Vitamin A

b) Vitamin K

c) Vitamin C

d) Vitamin D

7. Antidiuretic hormone controls reabsorption of [U] [CO:1]

a) Vitamins

b) Sodium

c) Glucose

d) Water

8. The junction between two neurons is called [U] [CO:2]

a) Synapse

b) Synaptic cleft

c) Axon

d) Dendron

9. Name the immunoglobulin which can cross the placenta [U] [CO:4]

a) IgG

b) IgA

c) IgM

d) IgE

10. The functional unit of a skeletal muscle [U] [CO:2]

a) Myofibril

b) Actin

c) Myosin

d) Sarcomere

11. Plasma cells are formed from [U] [CO:4]

a) T Lymphocytes

b) B Lymphocytes

c) Macrophages

d) Basophil

12. Blood transports CO_2 as [U] [CO:1]

a) Carboxyhaemoglobin

b) Haemoglobinic acid

c) Bicarbonates

d) Deoxyhaemoglobin

PART B

Short answer questions. Answer any 6. Each question carries 3 marks. (6 x 3= 18 marks)

13. Comment on the mechanism of action of steroid hormones. [U] [CO:3]

14. Explain Bohr Effect. [U] [CO:1]

15. What are neurotransmitters? Give an example. [U] [CO:2]

16. List the functions of the hormones produced by pancreas. [U] [CO:3]

17. Comment on DNA vaccines and Recombinant vaccines. [U] [CO:4]

18. Describe Cori Cycle. [U] [CO:2]

19. Comment on Carbon monoxide poisoning. [U] [CO:1]

20. Write briefly on monoclonal antibodies. [U] [CO:4]

PART C

Short essay questions. Answer any 4. Each question carries 5 marks. (4 x 5= 20 marks)

21. Explain the mechanism of impulse transmission across a synapse. [U] [CO:2]

22. Describe the transport of O_2 in blood. [U] [CO:1]

23. Explain the structure of a striated muscle. [U] [CO:2]

24. Comment on the hormones secreted by hypothalamus. [U] [CO:3]

25. Define hypersensitivity. What are the different types of hypersensitivity. [U] [CO:4]

26. Write an account on the cells of the immune system. [U] [CO:4]